Application Architecture – Smart Lender Project

Objective

Design a secure and scalable architecture that collects financial and customer data, processes it, predicts creditworthiness, and provides loan recommendations through a user-friendly interface.

High-Level Architecture

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                +----------------------+
                |   Web/Mobile Client  |
                +----------+-----------+
                           |
                           v
                +----------------------+
                |     API Gateway      |
                +----------+-----------+
                           |
        ------------------------------------------
         |                      |                  |
         v                      v                  v
+----------------+   +----------------+    +----------------+
| Authentication |   | Loan Service   |    | User Service   |
+----------------+   +----------------+    +----------------+
        |                       |                  |
        ------------------------------------------
```

```
                 |
                 v
          +----------------------+
          | Data Collection Layer|
          +----------+-----------+
                     |
     -------------------------------------------
          |              |                 |
          v              v                 v
  Banking APIs    Credit Bureau APIs    User Documents
                     |
                     v
          +----------------------+
          | Data Processing Layer|
          | Cleaning & Validation|
          +----------+-----------+
                    |
                    v
          +----------------------+
          | Feature Engineering  |
          +----------+-----------+
                     |
                     v
```

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+----------------------+
| ML Credit Scoring    |
+----------+-----------+
           |
           v
+----------------------+
| Recommendation Engine|
+----------+-----------+
           |
           v
+----------------------+
| Relational Database  |
+----------------------+
```

Architecture Components

Layer	Responsibilities
Presentation Layer	Web and mobile interfaces for customers and administrators.
API Gateway	Routes requests, authentication, rate limiting, and logging.
Business Logic Layer	Loan management, user management, eligibility checks, and notifications.
Data Collection Layer	Collects customer profile, banking history, credit reports, income details, and uploaded documents.
Data Processing Layer	Cleans, validates, transforms, and integrates collected data.

Layer	Responsibilities
Machine Learning Layer	Predicts credit score, default probability, and loan eligibility.
Recommendation Layer	Suggests suitable loan products and interest rates.
Database Layer	Stores customer information, transactions, loan applications, and ML outputs.

Data Sources

- Customer registration data
- Bank transaction history
- Credit bureau reports
- Income and employment information
- Government identity verification
- Existing loan records
- Mobile/app usage data (optional)
- Alternative credit data

Data Flow

1. User registers and submits loan application.
2. Customer data is collected from multiple sources.
3. Data validation and preprocessing are performed.
4. Cleaned data is stored in the database.

5. Feature engineering prepares inputs for the ML model.
6. The credit scoring model predicts risk.
7. The recommendation engine determines loan eligibility and suitable offers.
8. Results are displayed to the customer and loan officer.

Suggested Technology Stack

Layer	Technology
Frontend	React, Angular, Flutter
Backend	Spring Boot, Node.js, Django
API	REST APIs
Authentication	JWT, OAuth 2.0
Database	PostgreSQL, MySQL
NoSQL	MongoDB
Data Processing	Python, Pandas
Machine Learning	Scikit-learn, TensorFlow, XGBoost
Deployment	Docker, Kubernetes
Cloud	AWS, Azure, Google Cloud
Monitoring	Prometheus, Grafana

Security Considerations

- User authentication and authorization

- End-to-end encryption (TLS)
- Encryption of sensitive data at rest
- Role-based access control (RBAC)
- Audit logging
- Data privacy compliance (e.g., GDPR or applicable local regulations)
- API rate limiting and monitoring

Expected Deliverables for Epic 1

- Application architecture diagram
- Data flow diagram (DFD)
- Technology stack selection
- Database schema (high level)
- API architecture overview
- Data collection strategy
- Security architecture
- Non-functional requirements (performance, scalability, reliability, and availability)

This architecture provides a strong foundation for later epics involving machine learning model development, loan decision automation, and deployment of the Smart Lender system.